

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 –Coding Challenge	Assessment:	Assessment Tool 2 – Coding Challenge
Weightage:	40% of CIA	Total Marks:	100
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning (BL4)		
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Section / Batch:	B.E-CSE-AI	Date:	26-08-2026

Code of Conduct

I Shaik Nuha(192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk.Nuha

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Q. Logistic Regression with Interaction and Polynomial Terms on a Non-Linearly Separable Dataset

Objective

Demonstrate that:

1. Vanilla Logistic Regression fails on a non-linearly separable dataset (two concentric circles).
2. Feature Engineering using polynomial and interaction terms transforms the data into a linearly separable space.
3. Logistic Regression then achieves much better classification performance.

Theory

Dataset: Concentric Circles

Generate data:

```
from sklearn.datasets import make_circles
X, y = make_circles(
    n_samples=1000,
    noise=0.1,
    factor=0.4,
    random_state=42
)
```

The classes form:

- Inner Circle → Class 0
- Outer Circle → Class 1

A straight line cannot separate these classes.

Vanilla Logistic Regression

Decision boundary:

$$w_1 * x_1 + w_2 * x_2 + b = 0$$

Only linear boundaries are possible.

Result:

- Poor accuracy.
- Incorrect decision boundary.

Polynomial Feature Engineering

Create:

x_1^2

x_2^2

$x_1 * x_2$

The transformed model becomes:

$z =$

b

$+ w_1 * x_1$

$+ w_2 * x_2$

$+ w_3 * x_1^2$

$+ w_4 * x_2^2$

$+ w_5 * x_1 * x_2$

Now Logistic Regression can learn circular boundaries.

Complete Program

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_circles
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

```

# -----
# Generate Dataset
# -----

X, y = make_circles(
    n_samples=1000,
    noise=0.1,
    factor=0.4,
    random_state=42
)
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
)
# -----
# Vanilla Logistic Regression
# -----

lr = LogisticRegression()
lr.fit(X_train, y_train)
y_pred1 = lr.predict(X_test)
acc1 = accuracy_score(y_test, y_pred1)
print("Vanilla Logistic Regression Accuracy:")
print(acc1)
# -----
# Logistic Regression with
# Polynomial + Interaction Terms
# -----

```

```

poly_model = Pipeline([
    ("poly", PolynomialFeatures(
        degree=2,
        include_bias=False
    )),
    ("logistic", LogisticRegression())
])
poly_model.fit(X_train, y_train)
y_pred2 = poly_model.predict(X_test)
acc2 = accuracy_score(y_test, y_pred2)
print("\nPolynomial Logistic Regression Accuracy:")
print(acc2)
# -----
# Decision Boundary Plot Function
# -----
def plot_boundary(model, X, y, title):
    x_min, x_max = X[:,0].min()-0.5, X[:,0].max()+0.5
    y_min, y_max = X[:,1].min()-0.5, X[:,1].max()+0.5
    xx, yy = np.meshgrid(
        np.linspace(x_min, x_max, 500),
        np.linspace(y_min, y_max, 500)
    )
    Z = model.predict(
        np.c_[xx.ravel(), yy.ravel()]
    )
    Z = Z.reshape(xx.shape)
    plt.contourf(
        xx,yy, Z,
        alpha=0)

```

```

plt.scatter(
    X[:,0],
    X[:,1],
    c=y,
    edgecolors='k'
)
plt.title(title)
# -----
# Plot Results
# -----
plt.figure(figsize=(12,5))
plt.subplot(1,2,1)
plot_boundary(
    lr,
    X,
    y,
    f"Vanilla Logistic Regression\nAccuracy={acc1:.3f}"
)
plt.subplot(1,2,2)
plot_boundary(
    poly_model,
    X,
    y,
    f"Polynomial Logistic Regression\nAccuracy={acc2:.3f}"
)
plt.tight_layout()
plt.show()

```

Expected Results

Vanilla Logistic Regression

Accuracy typically:

0.45 – 0.60

Decision boundary:

Straight line

The model cannot separate the circles.

Polynomial Logistic Regression

Accuracy typically:

0.95 – 1.00

Decision boundary:

Circular / Non-linear

The model successfully separates the inner and outer circles.

Example Interpretation

Before Feature Engineering

Class 0 ○

Class 1 ◎

Logistic Regression:

Straight line

Cannot separate circles

After Feature Engineering

Using:

$$x_1^2$$

$$x_2^2$$

$$x_1 * x_2$$

The transformed space becomes linearly separable.

Logistic Regression learns:

$$x_1^2 + x_2^2 = c$$

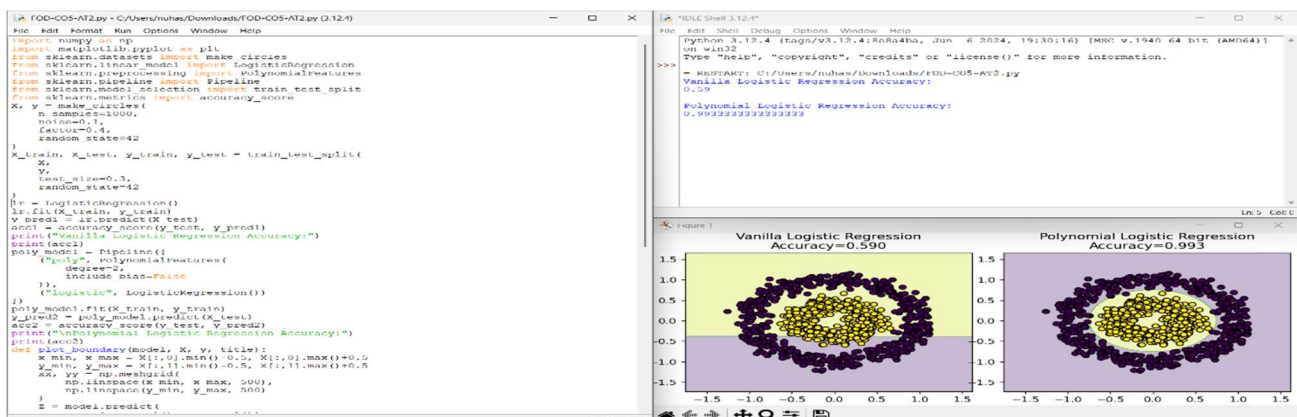
which corresponds to a circular decision boundary.

Conclusion

Model	Features Used	Boundary Type	Accuracy
Vanilla Logistic Regression	x_1, x_2	Linear	Low (~50%)
Logistic Regression + Polynomial Features	$x_1, x_2, x_1^2, x_2^2, x_1x_2$	Non-linear (circular)	High (~95–100%)

This experiment shows that logistic regression itself is a linear classifier, but with carefully engineered polynomial and interaction features, it can effectively model highly non-linear datasets such as concentric circles.

OUTPUT:



GEOTAG:

